Ryan Teck

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Second-year mechanical engineering student passionate about **innovation**, hands-on **problem solving**, and **robotics**. Seeking opportunities to apply my creativity and technical skills to tackle real-world engineering challenges.

EDUCATION

University of California San Diego — Expected June 2028

B.S. in Mechanical Engineering; *GPA: 3.9*

Relevant Coursework: Introduction to Mechanical Design, MATLAB Programming for Engineering Analysis, Thermodynamics, Elements of Materials Science, Statics and Introduction to Dynamics, Dynamics and Vibrations, Linear Circuits, Solid Mechanics, Signals and Systems

TECHNICAL SKILLS

CAD: SOLIDWORKS, Autodesk Fusion, AutoCAD
Programming: MATLAB, Python, Markdown

PROFESSIONAL EXPERIENCE

Research Project Contributor, Existential Robotics Lab — January 2025 – Present

- Authored accessible explanations of intelligent robotics systems (e.g., SLAM, localization) using Markdown and MkDocs, creating clear and digestible content for students of diverse backgrounds.
- Organized a workshop to teach mapping algorithms and created curated material for hands-on learning
- Collaborated with peers and mentors to conduct technical content reviews and optimize educational effectiveness.

Instructor, Kumon Learning Center — June 2023 – August 2024

- Tutored elementary through high school students in algebra, calculus, trigonometry, and advanced reading comprehension.
- Graded student assignments and provided individualized feedback to support their learning.

PROJECTS

Occupancy Grid Map, Existential Robotics Lab, Personal Project

- Implemented mapping algorithms utilizing a simulated LiDAR in a PyBullet simulation using Python
- Optimized code with methods such as vectorization, parallel processing, and profiling in NumPy
- Visualized a simulated car and map using plotting libraries such as OpenCV and Matplotlib

Robot Drivetrain, Class Final Project

- Designed a friction-drive drivetrain powered by DC motors using Autodesk Fusion and prepared parts for manufacturing using AutoCAD
- Machined parts using LaserCAMM, taps, drill presses, presses.
- Performed analyses to verify performance requirements and maintain appropriate factors of safety.
- Communicated with teammates to ensure components meshed properly and enabled efficient assembly.

Vice President, MESA (Math Engineering Science Achievement) — March 2023 - May 2024

- Coordinated 38 leadership members to host weekly meetings and events for 200+ members.
- Facilitated communication between USC, club advisors, and leadership.